

Beyond the Dots - A Database of Historical FOMC Dot Plots

Data Construction and Documentation

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Abstract

This document describes a dataset of individual Federal Open Market Committee (FOMC) participant projections for the federal funds rate recovered from the dot plots published in the Federal Reserve’s Summary of Economic Projections (SEP). The associated project provides an automated Python pipeline that downloads the official SEP documents, identifies the relevant dot-plot figure, detects the plotted participant projections, maps their graphical locations to federal funds rate values, and combines the extracted observations into a standardized data file. The pipeline covers the dot-plot era beginning in June 2012 and is designed to accommodate changes in the SEP graphical format through the present. This note documents the scope of the data, the principal extraction steps, the resulting data structure, and important considerations for users of the dataset.

1 Overview

The Federal Reserve publishes the [Summary of Economic Projections](#) in conjunction with selected [FOMC meetings](#). SEP participants report projections for major macroeconomic variables and their individual assessments of appropriate monetary policy. The monetary-policy projections are displayed as a dot plot, in which each dot represents an individual participant’s assessment of the appropriate target level, or midpoint of the target range, for the federal funds rate at the end of a specified calendar year or over the longer run. The Federal Reserve notes that these values are reported graphically and rounded to the nearest one-eighth percentage point in the modern SEP format.

The purpose of this project is to convert these graphical releases into a machine-readable longitudinal dataset. Rather than recording only the median or range published in the accompanying SEP tables, the dataset retains the distribution of the individual, anonymous participant projections represented by the dots. The resulting data can therefore be used to study the cross-sectional distribution, dispersion, central tendency, and evolution of FOMC participants’ policy-rate assessments over time.

The source code for the extraction pipeline is available at

<https://github.com/beyondthedots/beyondthedots.github.io/tree/main/code>,

and the data are available at:

<https://github.com/beyondthedots/beyondthedots.github.io/tree/main/data>,

2 Data Coverage

The dataset begins in **June 2012**, when the dot-plot format is available for the purpose of this extraction pipeline, and is designed to extend through the most recent SEP releases. The repository currently describes its coverage as **June 2012–September 2026**. SEP documents prior to 2012 are not included in the pipeline because the historical releases use numerical tables rather than the dot-plot format and consequently require a different extraction procedure.

The underlying source documents are the official SEP releases published by the Board of Governors of the Federal Reserve System. In the modern schedule, economic projections are generally collected and released four times per year, in connection with the March, June, September, and December FOMC meetings. The precise projection horizons vary across SEP releases as calendar years advance.

3 What the Project Contains

The repository is organized around an automated extraction workflow and its resulting output. The main components are:

- `run_2012-2014.py` and `run_2015-plus.py`: Python scripts and supporting code used to obtain and process SEP documents and extract dot-plot observations;
- separate extraction logic for the **2012–2014** and **2015 onward** graphical formats;
- a master script, `run_all_seps.py`, that executes the historical and modern extraction pipelines; and
- `data/projections.csv`: the consolidated machine-readable dataset produced by the pipeline.

The repository documentation reports more than 10,000 extracted projection observations in the combined output. The extraction is designed to operate directly from Federal Reserve SEP documents rather than from manually transcribed dot-plot values.

4 Data Structure

The consolidated output records an observation for each extracted dot. Its core variables are shown in Table 1.

Table 1: Core variables in the extracted dataset

Variable	Description
Year	Projection horizon associated with the dot (for example, 2025, 2026, or a corresponding coded horizon when applicable).
Projection	Extracted federal funds rate projection, in percentage points.
SEP_Date	Date of the SEP release/FOMC meeting associated with the projection.

Multiple rows can have identical values of **Year**, **Projection**, and **SEP_Date**. This is intentional: several participants may place dots at the same projected rate. Consequently, the multiplicity

of rows at a given rate and horizon contains information about the distribution of participant assessments and should generally be preserved.

The dots are anonymous. The dataset therefore identifies the distribution of individual projections within an SEP release but does *not* link a particular projection to a named FOMC participant, nor does it track an individual’s projections across meetings or horizons.

5 Data Construction Pipeline

The extraction procedure can be summarized in six principal steps.

A. Obtain the official SEP documents

For each relevant FOMC projection date, the pipeline retrieves the corresponding SEP PDF from the Federal Reserve website. The date of the source document is retained and later becomes the `SEP_Date` field in the consolidated data.

B. Locate the dot-plot figure

The page containing the monetary-policy dot plot is identified within each SEP document. Because the structure of the SEP changed over time, the relevant page is not fixed across the full sample. The modern pipeline accounts for these changes, including the shift in the typical location of the figure between earlier and later SEP documents.

C. Convert the figure to an image and isolate graphical elements

The relevant PDF page is converted into an image suitable for computer-vision processing. Image-processing operations are then used to isolate the plotted dots and distinguish them from axes, grid lines, labels, and other graphical elements. The implementation uses OpenCV-based image processing and format-specific detection rules.

D. Detect individual dots

Candidate dot locations are identified from the processed image. The implementation differs across historical formats. For the 2012–2014 documents, the pipeline uses distance-transform-based detection tailored to the tighter early dot-plot design. For the 2015-and-later documents, it uses an erosion-based detection procedure together with color information. Year-specific HSV color ranges are used where needed to accommodate changes in the appearance of the Federal Reserve figures.

When several participants report the same projected rate for the same horizon, dots are vertically stacked in the published figure. The extraction procedure is designed to recover the multiplicity represented by these stacked dots rather than treating a stack as a single observation.

E. Assign dots to projection horizons and rate values

Detected dots must be mapped from image coordinates to economic quantities. Horizontally, the pipeline groups dot locations into projection-year columns using clustering, including K-means-based year assignment. Vertically, grid-line detection is used to calibrate image position to the corresponding federal funds rate.

This calibration is format dependent. The repository distinguishes the early 2012–2014 figures from the 2015-and-later figures because grid spacing, plotting increments, and graphical conventions changed over time. The historical pipeline includes special corrections for early 2012 releases and for the transition in late 2014. The modern pipeline uses the finer grid structure of later SEP figures and maps detected positions to the corresponding projection increments.

F. Standardize and combine the extracted observations

After the dots have been assigned to horizons and rate values, observations from all SEP releases are placed into a common tabular format. The historical and modern outputs are combined and written to `data/projections.csv`. Running

```
cd code/  
python3 run_all_seps.py
```

executes the two extraction pipelines and produces the consolidated file.

6 Format Changes Across the Sample

A single fixed image-processing rule is not adequate for the entire history of the SEP dot plot. The project therefore explicitly separates two broad graphical regimes.

2012–2014. The early dot plots require tighter dot detection and use different grid conventions. The code uses distance-transform-based detection, accounts for the early grid spacing and projection increments, applies an offset correction to early 2012 dates, and contains special handling for the September/December 2014 transition in graphical spacing.

2015 onward. The later pipeline uses erosion-based dot detection, finer grid calibration, automatic handling of changes in the PDF page on which the dot plot appears, and HSV-based color detection with year-specific ranges. These adjustments are intended to make the extraction robust to changes in the Federal Reserve’s graphical presentation while retaining a common output format.

7 Interpretation and Limitations

The data should be interpreted as a machine-readable reconstruction of the anonymous individual policy-rate projections displayed in the official SEP dot plots. A dot represents a participant’s assessment of appropriate monetary policy conditional on that participant’s economic outlook; it is not necessarily a forecast of the policy decision the FOMC will ultimately make. The Federal Reserve also emphasizes that SEP projections are individual participant assessments rather than a collectively agreed Committee forecast.

Because the source is graphical, the historical data are obtained through image extraction rather than from participant-level numerical records. Changes in image resolution, colors, page layout, grid spacing, and overlapping or stacked graphical elements can therefore affect extraction. The pipeline includes date- and format-specific logic to address these issues, but users conducting applications in which individual observations are especially consequential should validate the relevant SEP releases against the original Federal Reserve figures.

The data do not reveal participant identities. They consequently support analysis of the distribution of policy projections at a given meeting but not participant-level panel analysis. Finally, pre-2012 projections are outside the present scope because those releases require extraction from numerical tables rather than dot plots.

Disclaimer

Beyond the Dots is an independent research project and is not affiliated with, endorsed by, or sponsored by the Board of Governors of the Federal Reserve System or any Federal Reserve Bank. The data are compiled from publicly available sources and are provided for research, educational, and informational purposes.

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